

Efficacy and Safety of Different Add Power Soft Contact Lenses on Myopia Progression in Children: A Systematic Review and Meta-Analysis

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Keywords

Myopia · Children · Soft multifocal contact lenses · Efficacy · Safety

Abstract

Background: In children, myopia has become a widespread and serious global public health problem. Soft multifocal contact lenses (SMCLs) have been widely studied to control myopia progression in children. However, their efficacy in myopia control in children and its adverse effects and which added power SMCLs are more effective and safer remains to be explored. **Objectives:** Evaluate the efficacy and safety of various add power SMCLs to slow myopia progression in children. **Method:** Eligible randomized controlled trials were retrieved from PubMed, MEDLINE, EMBASE, and the Cochrane Central Register of Controlled Trials databases. The present meta-analysis analyzed the mean differences (MD) in myopic progression, axial length, and odds ratios for adverse effects and dropout rates between SMCLs with different added powers and control groups. Changes in visual performance were also systematically evaluated. **Results:** Seven independent studies involving 805 children were included in the present meta-analysis. At 1-year, the weighted MD

(WMD) in myopia progression between SMCL and control groups was -0.22 diopters (D) (95% confidence interval [CI]: -0.56 to 0.12 D) for low add power SMCLs, 0.09 D (95% CI: $0-0.19$ D) for medium add power SMCLs, and 0.2 D (95% CI: $0.13, 0.27$ D) for high add power SMCLs. At 2-years, the WMD for medium add power was 0.12 D (95% CI: -0.03 to 0.27 D), and for high add power was 0.25 D (95% CI: $0.14-0.35$ D). No differences were detected for adverse effects ($p = 0.2$) and acceptability ($p = 0.74$) between different added powers. Additionally, differences in visual performance changes, produced by different added powers, were not detected. **Conclusions:** The present meta-analysis showed that high add power SMCLs are more effective and stable to control myopia progression. Besides, the adverse effects and acceptability were not related to the added power.

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Introduction

Recent studies have indicated that myopia affects a large proportion of the world's population and has become a widespread and increasing public health concern.

Approximately, 312 million cases of myopia were recorded worldwide in 2015. In East Asia, the highest prevalence was among children (0–19 years), reaching 69% at 15 years (86% among Singaporeans-Chinese) [1]. About 49.8% (4.758 billion people) and 9.8% (938 million people) of the global population will have myopia and high myopia in 2050, respectively [2], including 84% of Chinese children [3]. Similarly, the prevalence of myopia is about 40% in Europe and 42% in North America [4]. After the social isolation promoted by COVID-19, the prevalence of myopia in children with 6–8 years increased 1.4–3 times [5]. The myopia progression in European children, including Turkey and Spain, was also significantly faster than before the pandemic [6]. Therefore, myopia prevention and control among children has to be strengthened to prevent an increase in the number of patients worldwide. Myopia, especially high myopia, might be complicated by visual function damages, such as retinal detachment, primary open-angle glaucoma, cataracts, and macular degeneration. According to a previous study, 10 million people had visual impairment from myopic macular degeneration in 2015 (0.13% prevalence), of which 3.3 million were blind (0.04% prevalence) [7]. Without changes to current interventions, by 2050, the population visually impaired by myopic macular degeneration is projected to reach 55.7 million people (0.57% prevalence), with 18.5 million blind (0.19% prevalence) [8]. Moreover, myopia has become one of the leading causes of irreversible blindness and visual impairment, comprehending a big economic burden of approximately \$709 per person per year [9]. Therefore, effective treatments to slow or even stop myopia progression in young children are urgent. Recently, researchers and clinicians have proposed different treatment strategies, such as keratoplasty and low-concentration atropine. However, until now, no method can effectively prevent and treat myopia with sufficient safety and acceptability.

SMCLs provide multiple focal planes or simultaneous distance corrections and retinal myopic defocus [10–12]. Although their exact mechanism of action is still unknown, SMCLs with different add powers have been widely used to control myopia progression in children [13–18], especially in Asia, and are also of interest in Europe and North America. SMCLs effects are thought to be related to the added powers and potentially cause other visual function reductions. Previous studies have found that high add power SMCLs are more effective in slowing myopia progression, but some subjects presented alterations in visual acuity, contrast sensitivity, and regulatory function [19–21]. Additionally, recent studies have found

that SMCLs with add power of +0.5 diopters (D) are effective in controlling myopia progression and have less impact on visual acuity [22]. Thus, to date, there are many uncertainties related to the clinical use of SMCLs, such as the add power size, safety concerns and acceptability, and whether there is variability between different ethnic groups and SMCLs designs.

Previous systematic reviews and meta-analyses have assessed the effectiveness of SMCLs but did not explore differences in efficacy across add powers and perform a quantitative assessment of adverse effects and acceptability. In the present follow-up meta-analysis, we analyzed overall differences in efficacy, adverse effects, and clinical acceptability of SMCLs with different add powers to slow myopia progression in children. Also, we explored whether differences existed between different ethnic groups and SMCLs designs. Finally, we investigated changes in visual functions such as visual acuity, contrast sensitivity, and regulatory function.

Methods

Registration and Protocol Statement

Registration number: CRD42020208256. <https://www.crd.york.ac.uk/PROSPERO/#recordDetails>.

Eligibility Criteria

Studies were selected based on the following criteria: (1) study type: randomized controlled trials (RCTs); (2) participants: school-age children (6–18 years) with myopia; (3) intervention: SCL experimental groups in a multifocal design with specific add powers and monovision soft contact lenses or glasses in control groups; and (4) outcomes: myopic progression (change in spherical equivalents) as the primary outcome and axial elongation (axial length changes) as the secondary outcome. Two authors (X.Z. and D.L.) independently assessed the titles and abstracts of all retrieved studies.

Search Strategy

We searched the PubMed, MEDLINE, EMBASE, and the Cochrane Central Register of Controlled Trials databases for relevant studies from inception to July 10, 2021, using medical subject headings (MeSH) and free words combining myopia, contact lens, and RCTs. We also screened Clinicaltrials.gov and the reference list of published reviews to identify additional relevant studies. The Boolean operators “AND,” “OR,” and “NOT” were used to combine all search sets.

Data Extraction

Two authors (Z.Y. and A.Z.) screened titles and abstracts to independently and repeatedly identify potentially eligible articles. Then, they examined the full text to determine the final inclusion. For included studies, two reviewers independently extracted study characteristics (authors, year, country or region, study design, follow-up time, interventions and controls, added power), patient

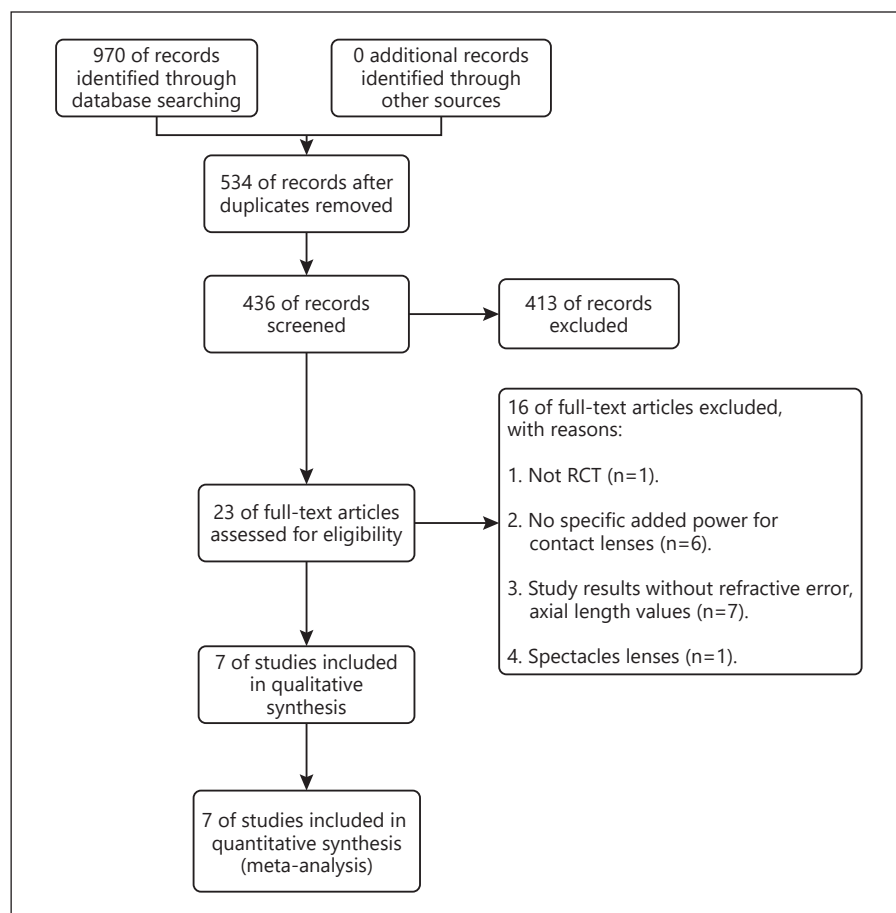


Fig. 1. Flow diagram of literature search.

characteristics (age, mean change in cycloplegic spherical equivalents, mean axial elongation changes, number of adverse events and dislocations), and relevant information regarding the methods. For continuous data, SER and axial length change values were included and are expressed as means and SDs. For categorical data, the number of each group was extracted. For missing data, the Get-Data Graph Digitizer 2.25 (<http://getdata-graph-digitizer.com>) was used for figures illustration only, if needed. Discrepancies were adjudicated by a third reviewer (X.Z.).

Quality Assessment

The risk of bias for the selected trials was assessed according to six aspects of the Cochrane Collaboration: sequence generation, allocation concealment, blinding, incomplete outcome information, selective outcome reporting, and other sources of bias. The risk of bias for each aspect was graded as “low,” “unclear,” or “high.”

Statistical Analyses

Data analyses were performed using review manager (version 5.4; Cochrane Collaboration) and STATA (version 16.0; Stata-Corp). We performed a comparative study analysis based on the changes in contact lenses with different added powers versus control groups. We calculated the weighted mean difference (WMD) and 95% confidence intervals (CIs) for different contact lens add-

ed powers in refractive changes and axial elongation versus control groups. We also analyzed the odds ratio for adverse effects and the number of dislocations between SMCLs and control groups and presented them as forest plots. Statistical heterogeneity between studies was tested using the χ^2 test, Tau^2 , and I^2 , and a $p < 0.1$ was considered significant. Larger Tau^2 and I^2 indicated greater heterogeneity. If significant heterogeneity existed, estimates were calculated using a random-effects model. Otherwise, a fixed-effects model was used. In this paper, sensitivity analysis was conducted by excluding literature with significantly different characteristics. Subgroup analyses were performed by heterogeneity across ethnicity (Asian vs. White individuals) and contact lens designs. Since <10 trials were included, funnel plots were used to check for publication bias and to analyze the symmetry.

Results

Selection and Characteristics of Studies (Online Suppl. Tables 1–6)

In the initial search, 970 studies were identified. Then, 534 duplicates, 412 irrelevant studies, and 17 studies that assessed other outcomes without exact added powers and

Table 1. Characteristics of the studies included in the meta-analysis

Study (author, year)	Country or area	Study design	Follow-up, months	Age (years)	Classification (i/ii/iii)	Plus power	Total No of patients (E/C)	Baseline refraction (D)		Intervention	
								experimental	control	experimental	control
Anstice and Phillips [28] 2011	New Zealand	RCT (crossover)	10	11–14	i	+2.00 D	35/35	–2.71±1.10	–2.71±1.10	Dual-focus soft contact lens	Single-vision contact lenses
Lam et al. [26] 2014	Hong Kong	RCT	12/24	8–13	i	+2.50 D	65/63	–2.90±1.05	–2.80±1.03	Defocus incorporated soft contact	Single-vision contact lenses
Fujikado et al. [22] 2014	Japan	RCT	12	10–16	iii	+0.50	24/24	–2.56±0.87 D	–2.64±0.99 D	Progressive-addition power	Single-vision contact lenses
Ruiz-Pomeda [25] 2018	Spain	RCT	12/24	8–12	ii	+2.00 D	41/33	–2.16±0.94 D	–1.75±0.94 D	MiSight CLs	Single-vision spectacle
Sankaridurg [24] 2019	China	RCT	24	8–13	iii	+2.50	103/102	–2.39±0.79	–2.29±0.75	Central and peripheral plus contact lenses	Single-vision contact lenses
Walline [23] 2020	USA	RCT	36	7–11	ii	+2.50	97/97	–2.28±0.90	–2.46±0.97	Soft multifocal contact lenses	Single-vision contact lenses
						+1.50	98/97	–2.43±1.11	–2.46±0.97		
Garcia-Del Valle et al. [27] 2021	Spain	RCT	12	7–15	iii	+2.00 D	32/26	–2.80±1.79	–3.31±1.76	Progressive multifocal CLs	Placebo

(i) Bifocal contact lenses. (ii) Multifocal contact lens. (iii) Progressive power contact lens/varifocal power contact lens. E, Experimental; C, control; CLs, contact lenses.

Table 2. Quality assessment of included RCTs in the meta-analysis

Studies	Items						
	random sequence generation	allocation concealment	blinding of performance and detection	incomplete outcome data	selective reporting	other bias	grade of risk bias
Anstice and Phillips [28] 2011	L	U	L	L	L	U	A
Lam et al. [26] 2014	L	U	L	L	L	U	A
Fujikado et al. [22] 2014	L	U	L	L	L	U	A
Ruiz-Pomeda et al. [25] 2018	L	L	L	U	L	U	A
Sankaridurg et al. [24] 2019	L	L	L	L	L	U	A
Walline et al. [25] 2020	L	L	L	L	L	U	A
Garcia-Del Valle et al. [27] 2021	L	U	L	L	L	U	A

A: 4 and more items meeting the “L” level; B: 2 or 3 items meeting the “L” level; C: 1 or no item meeting the “L” level or 1 and more meeting the “H” level. L, low (low risk bias); H, high (high risk bias); U, unclear.

with baseline information only were excluded. Finally, seven independent studies were included (Fig. 1) [22–28]. A total of 805 children younger than 18 were included in this meta-analysis, including 346 Asians and 459 White individuals. Studies’ characteristics are listed in Table 1. One study [22] of low added power contact lenses, two [23, 24] of medium added power contact lenses, and six [23–28] of high added power contact lenses were included, yielding nine experimental groups in seven studies. Two studies [25, 27] were conducted in Spain, one in China [24], one in Hong Kong [26], one in the USA [23], one in New Zealand [28], and one in Japan [22]. Anstice and Phillips [28] performed their self-controls and compared interocular imbalances. The other studies included SMCLs versus a control group with SCL or single-vision spectacle.

PRISMA Checklist

The PRISMA Guidelines are provided in online supplementary Table 7 (for all online suppl. material, see www.karger.com/doi/10.1159/000523675).

Risk of Bias Assessment

All trials reported adequate randomization methods, the number of participants lost to follow up, and completeness of outcome indicators, comprehending a low risk of bias. All trials reported blinding, with five studies blinded to outcome measures and two blinded to subjects and investigators, leading to a low risk of bias. Four studies did not describe allocation protocol concealment and had some risk of bias. Seven studies had other unclear biases, comprehending some risk of bias. Overall, the quality of the RCTs included was high (Table 2).

Refraction

In the present study, data were analyzed separately for 1- and 2-year refractive error change values (Table 3). When the 1-year SMCLs were compared to controls, one study involved low add power contact lenses; two involved medium add power contact lenses, and six studies reported data for high add power contact lenses. At 1 year, the WMD for low add power was -0.22 D (95% CI: -0.56 to 0.12 D; $p = 0.21$), for medium add power was 0.09 D (95% CI: 0 – 0.19 D; $p = 0.06$), and for high add power was 0.20 D (95% CI: 0.13 – 0.27 D; $p < 0.001$) (Fig. 2). When the 2-year SMCLs were compared to controls, two studies reported data for medium add power contact lenses and four reported data for high add power contact lenses. The WMD for medium add power was 0.12 D (95% CI: -0.03 to 0.27 D; $p = 0.10$) and for high add power was 0.25 D (95% CI: 0.14 – 0.35 D; $p < 0.001$) (Fig. 3). No significant heterogeneity was detected and all data were analyzed using a fixed-effects model. Significant differences were detected between the 1-year different added power SMCLs groups ($p = 0.02$), with significant efficacy in the high add power SMCLs group compared to controls ($p < 0.001$) and no significant efficacy in the low add power and medium add power SMCLs groups ($p = 0.21$ and $p = 0.06$, respectively). No significant differences were detected between the 2-year different added power groups ($p = 0.17$). Compared to controls, the high add power SMCLs group had significant efficacy ($p < 0.001$) and the medium add power SMCLs group had no significant efficacy ($p = 0.10$). No significant publication bias was found for all trials using the funnel plot (Fig. 4). Moreover, no significant differences were found between

Table 3. Myopia progression and axial elongation of included studies during different periods of follow-up

Studies (author, year)	Follow-up, months	Myopia progression, D		Axial elongation, mm	
		experimental	control	experimental	control
<i>Low-addition soft contact lenses</i>					
Fujikado et al. [22] 2014	12	0.84±0.42	0.62±0.43	0.15±0.07	0.20±0.09
<i>Medium addition soft contact lenses</i>					
Sankaridurg et al. [24] 2019	6	0.27±0.31	0.39±0.25	0.12±0.11	0.19±0.09
	12	0.53±0.43	0.66±0.33	0.24±0.17	0.33±0.14
	18	0.70±0.48	0.88±0.40	0.35±0.24	0.47±0.21
	24	0.87±0.56	1.12±0.51	0.45±0.29	0.58±0.27
Walline et al. [23] 2020	12	0.36±0.51*	0.41±0.46*	−0.01±0.24*	−0.03±0.24*
	24	0.66±0.83*	0.67±0.60*	0.59±0.31*	0.61±0.27*
	36	0.86±0.78*	1.01±0.62*	0.57±0.31*	0.65±0.27*
<i>High-addition soft contact lenses</i>					
Anstice and Phillips [28] 2011	10	0.44±0.33	0.69±0.38	0.11±0.08	0.22±0.09
Lam et al. [26] 2014	6	0.21±0.46*	0.26±0.51*	0.07±0.21*	0.11±0.23*
	12	0.36±0.46*	0.48±0.47*	0.13±0.21*	0.21±0.24*
	18	0.50±0.59*	0.71±0.61*	0.20±0.23*	0.29±0.27*
	24	0.59±0.60*	0.80±0.66*	0.25±0.24*	0.36±0.28*
Ruiz-Pomeda et al. [25] 2018	12	0.18±0.54*	0.43±0.42*	0.12±0.13*	0.24±0.35*
	24	0.45±0.62*	0.73±0.58*	0.28±0.27*	0.44±0.27*
Walline et al. [23] 2020	12	0.22±0.45*	0.41±0.46*	0±0.22*	−0.03±0.24*
	24	0.38±0.60*	0.67±0.60*	0.38±0.27*	0.61±0.27*
	36	0.54±0.64*	1.01±0.62*	0.39±0.26*	0.65±0.27*
Sankaridurg et al. [24] 2019	6	0.25±0.29	0.39±0.25	0.11±0.11	0.19±0.09
	12	0.50±0.35	0.66±0.33	0.21±0.17	0.33±0.14
	18	0.71±0.53	0.88±0.40	0.34±0.25	0.47±0.21
	24	0.92±0.56	1.12±0.51	0.44±0.29	0.58±0.27
Garcia-Del Valle et al. [27] 2021	6	0.25±0.32	0.46±0.53	0.07±0.10	0.09±0.10
	12	0.28±0.35	0.57±0.52	0.13±0.12	0.22±0.14

The myopia control rate is defined as the ratio of the difference between the two groups to the difference in the control group. * SD, from other studies used the same measurement scale, had the same degree of measurement error, had the same time interval between baseline and post-intervention measurement, and in a similar population.

$$SD_{E, \text{change}} = \sqrt{SD_{E, \text{baseline}}^2 + SD_{E, \text{final}}^2 - (2 \times \text{Corr} \times SD_{E, \text{baseline}} \times SD_{E, \text{final}})}.$$

Asians and White individuals for SMCLs treatments at 1 and 2 years ($p = 0.31$ and $p = 0.77$, respectively) (Fig. 5–6). Also, no significant differences were found between bifocal, multifocal, and progressive power contact lenses for SMCLs treatments at 1 and 2 years ($p = 0.84$ and $p = 0.81$, respectively) (Fig. 7–8).

Axial Elongation

The studies reporting axial length data were consistent with refraction (Table 3). When the 1-year SMCLs was

compared to controls, the WMD for low add power was 0.05 mm (95% CI: −0.11 to 0.01 mm; $p = 0.13$), for medium add power was 0.04 mm (95% CI: −0.14 to 0.07 mm; $p = 0.50$), and for high add power was 0.08 mm (95% CI: −0.13 to 0.03 mm; $p = 0.0007$) (Fig. 9). When the 2-year SMCLs were compared to controls, the WMD for medium add power was 0.07 mm (95% CI: −0.17 to 0.04 mm; $p = 0.21$) and for high add power was 0.17 mm (95% CI: −0.22 to −0.11 mm; $p < 0.001$) (Fig. 10). No significant differences were detected between 1-year ($p = 0.64$) and

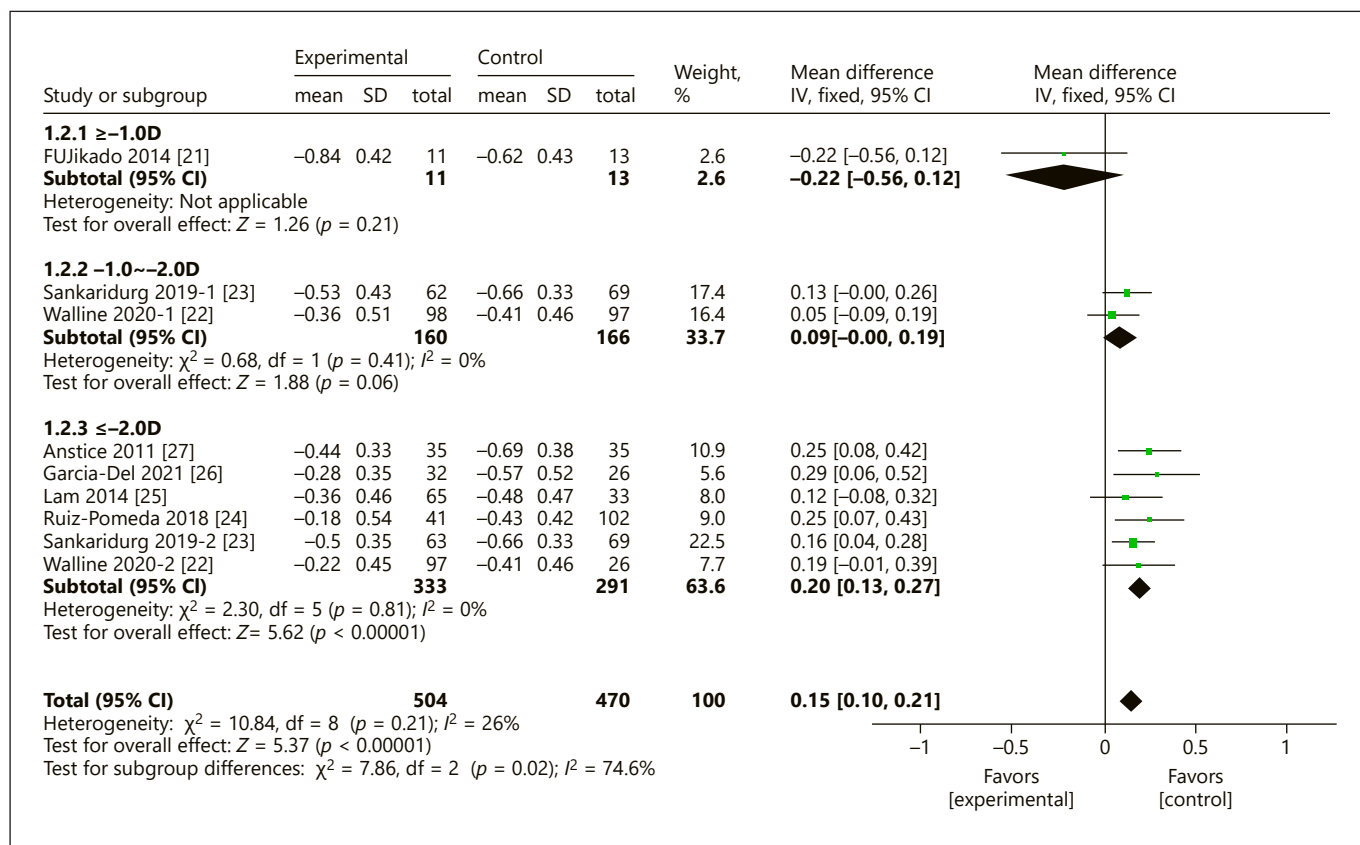


Fig. 2. Forest plots of SE changes in different contact lens additions at 1 year.

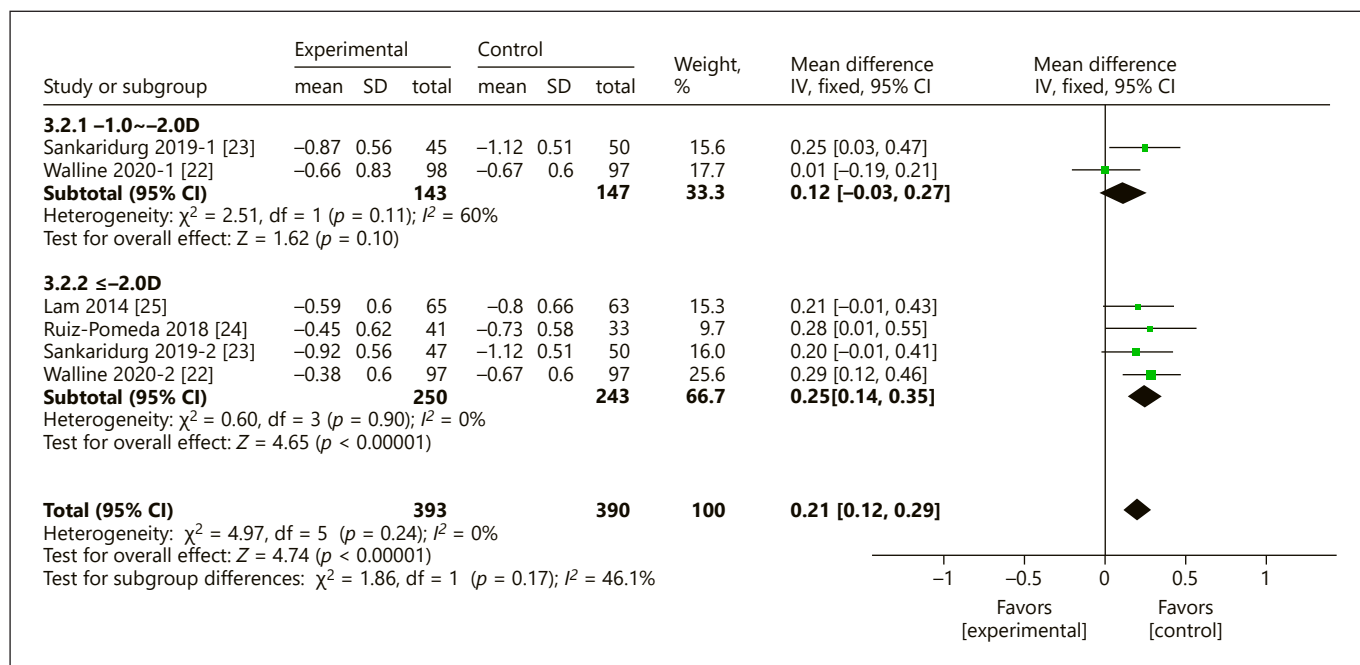


Fig. 3. Forest plots of SE changes in different contact lens additions at 2 years.

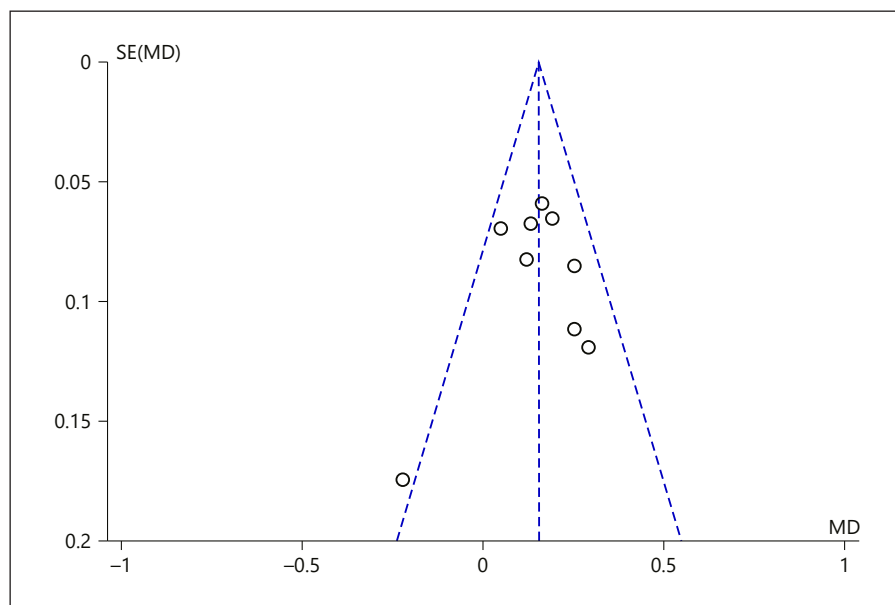


Fig. 4. Funnel plot of SE.

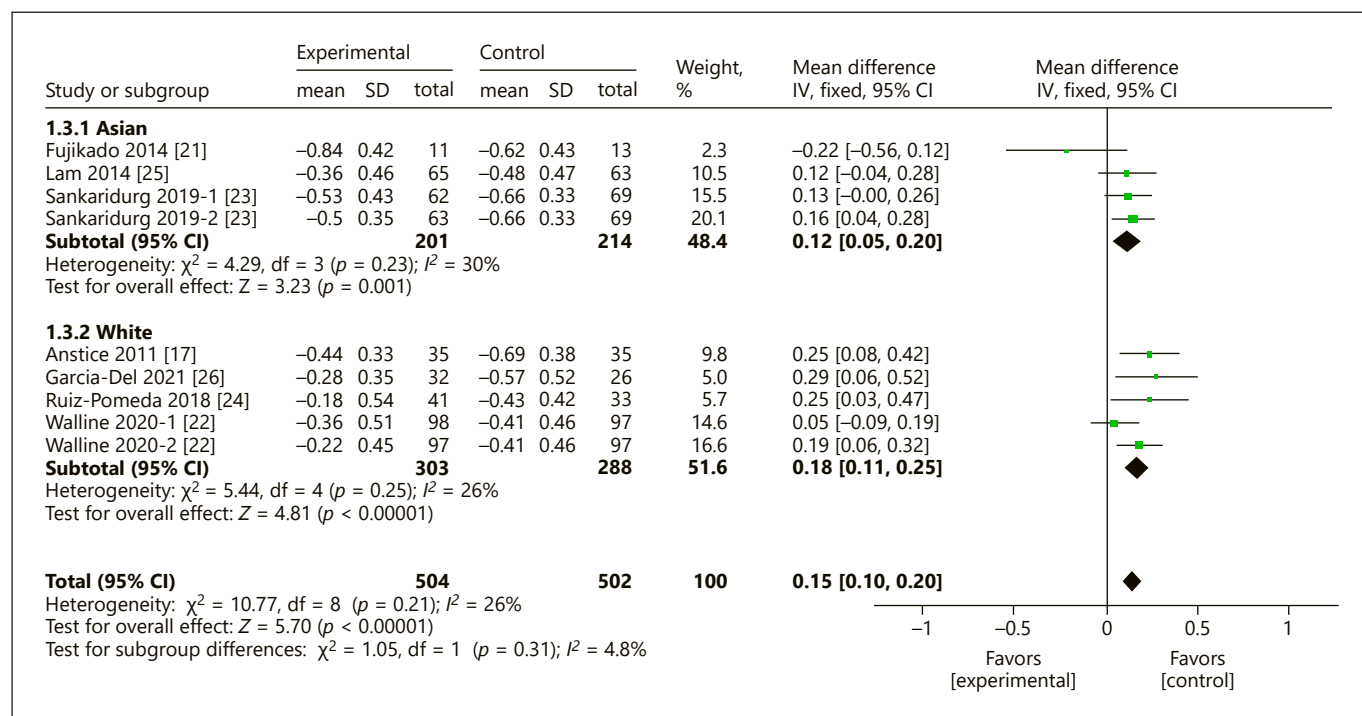


Fig. 5. Forest plots of SE changes in different ethnic groups at 1 year.

2-year ($p = 0.12$) SMCLs groups with different added powers. Additionally, no significant efficacy was found for the low add power SMCLs groups compared to controls ($p = 0.13$). Heterogeneity was found in all axial

length analyses (Tables 4, 5), thus a random-effects model was used. By sensitivity analysis (excluded one study at a time), heterogeneity disappeared after excluding two studies, Walline et al. [23] 2020 at 1-year ($\text{Tau}^2 = 0$, $I^2 =$

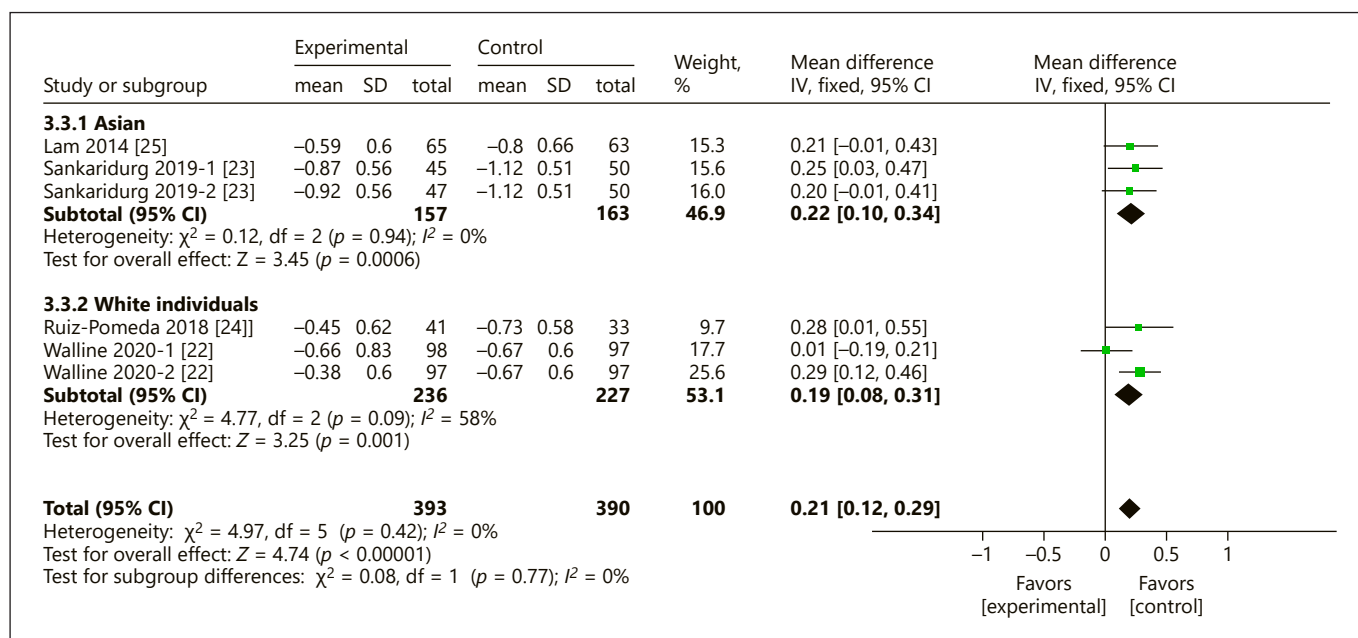


Fig. 6. Forest plots of SE changes in different ethnic groups at 2 years.

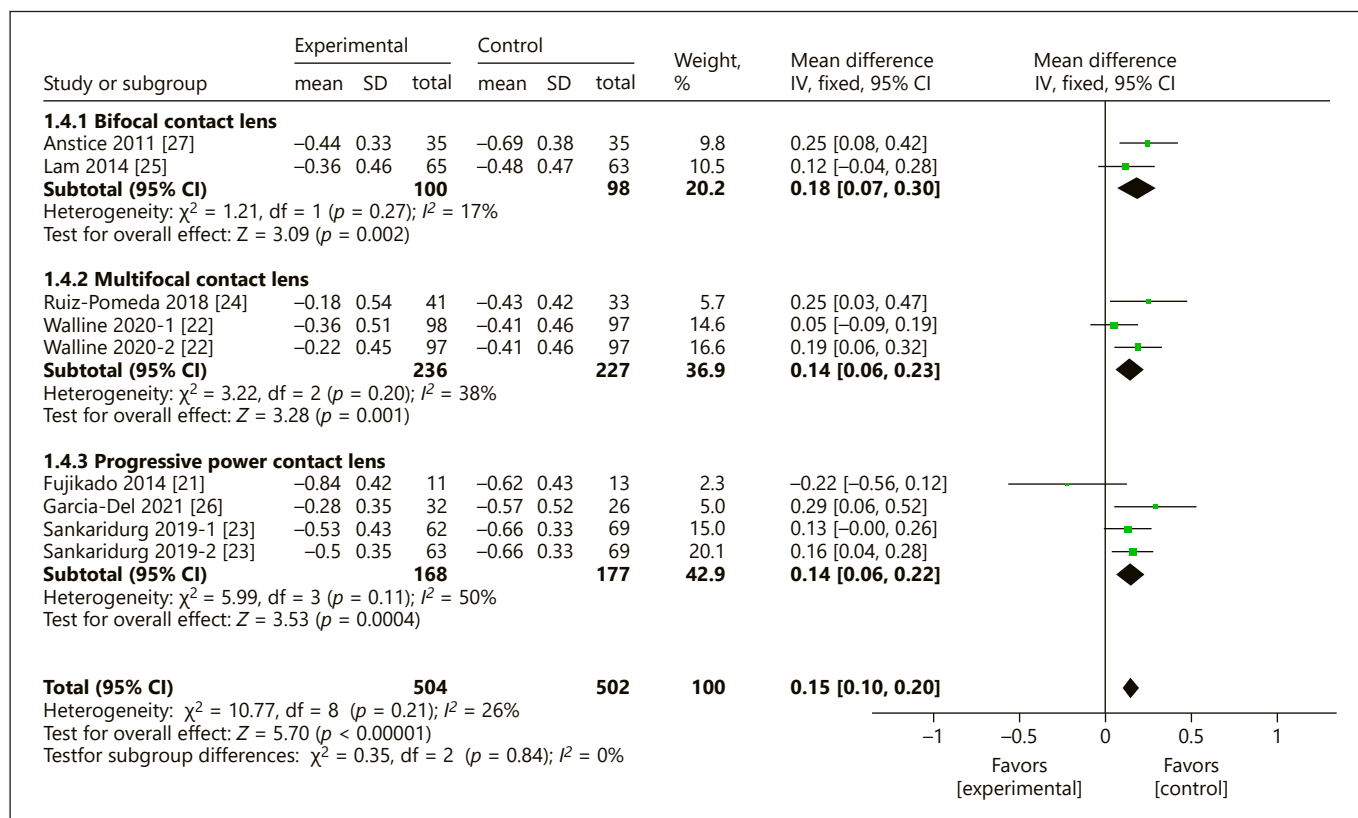


Fig. 7. Forest plots of SE changes for different contact lens design types at 1 year.

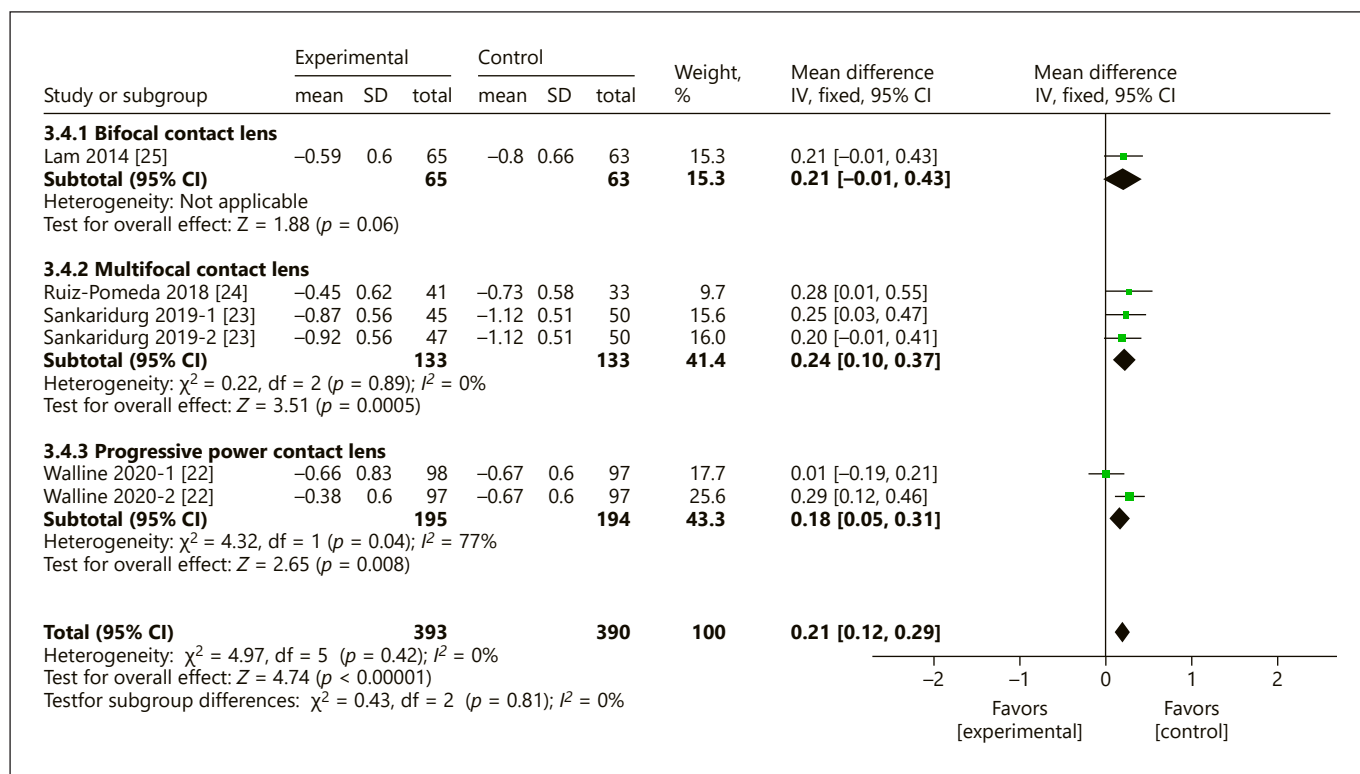


Fig. 8. Forest plots of SE changes for different contact lens design types at 2 years.

0%, $p = 0.73$), and decreased after excluding one study, Walline et al. [23] 2020 at 2-years ($\text{Tau}^2 = 0.00$, $I^2 = 17\%$, $p = 0.31$). The overall effect of the studies and the difference in effect between subgroups was not reversed (Tables 4, 5). No significant publication bias was found for all trials by the funnel plot (Fig. 11). No significant differences were found between Asians and White individuals for SMCLs treatments at 1 and 2 years ($p = 0.46$ and $p = 0.05$, respectively) (Fig. 12–13). Bifocal, multifocal, and progressive power contact lenses also did not reveal significant differences between 1-year and 2-year SMCLs treatments ($p = 0.41$ and $p = 0.10$, respectively) (Fig. 14–15).

Adverse Effects

Three trials from two studies reported adverse effects with no significant difference in the incidence of adverse reactions between different add power SMCLs ($p = 0.2$; Fig. 16). The adverse events from the three trials are summarized in Table 6. A total of 32 adverse events were reported in 235 patients in the SMCLs group, comprehending an incidence of 12.8%. On the other hand, 17 adverse events were reported in 127 patients in the control group,

comprehending an incidence of 13.4%. The most common adverse effects were papillary conjunctivitis ($n = 10$, 2.8%), infiltrative keratitis ($n = 8$, 2.2%), and ocular allergy ($n = 7$, 1.9%). Other adverse effects ($\leq 1\%$ of the patients) included micropapillary response, corneal neovascularization, superficial punctate keratitis, corneal epithelial defect/erosion, contact lens associated red-eye, sterile corneal ulcer, toxic gas, soap in the eye, superior epithelial arcuate lesion, and probable microbial keratitis. The three studies reporting adverse effects involved White individuals and the SMCLs design was of progressive power contact lens.

Visual Performance

Eight experiments from six studies reported changes in visual performance, including contrast visual acuity, visual acuity, and accommodation. Three studies reported contrast visual acuity, including one reporting lower high and low contrast visual acuity in SMCLs than in controls, one reporting lower low contrast visual acuity in the SMCLs group without clinical significance, and one with no differences. Two studies reported visual acuity, with no differences between SMCLs and controls. Two studies

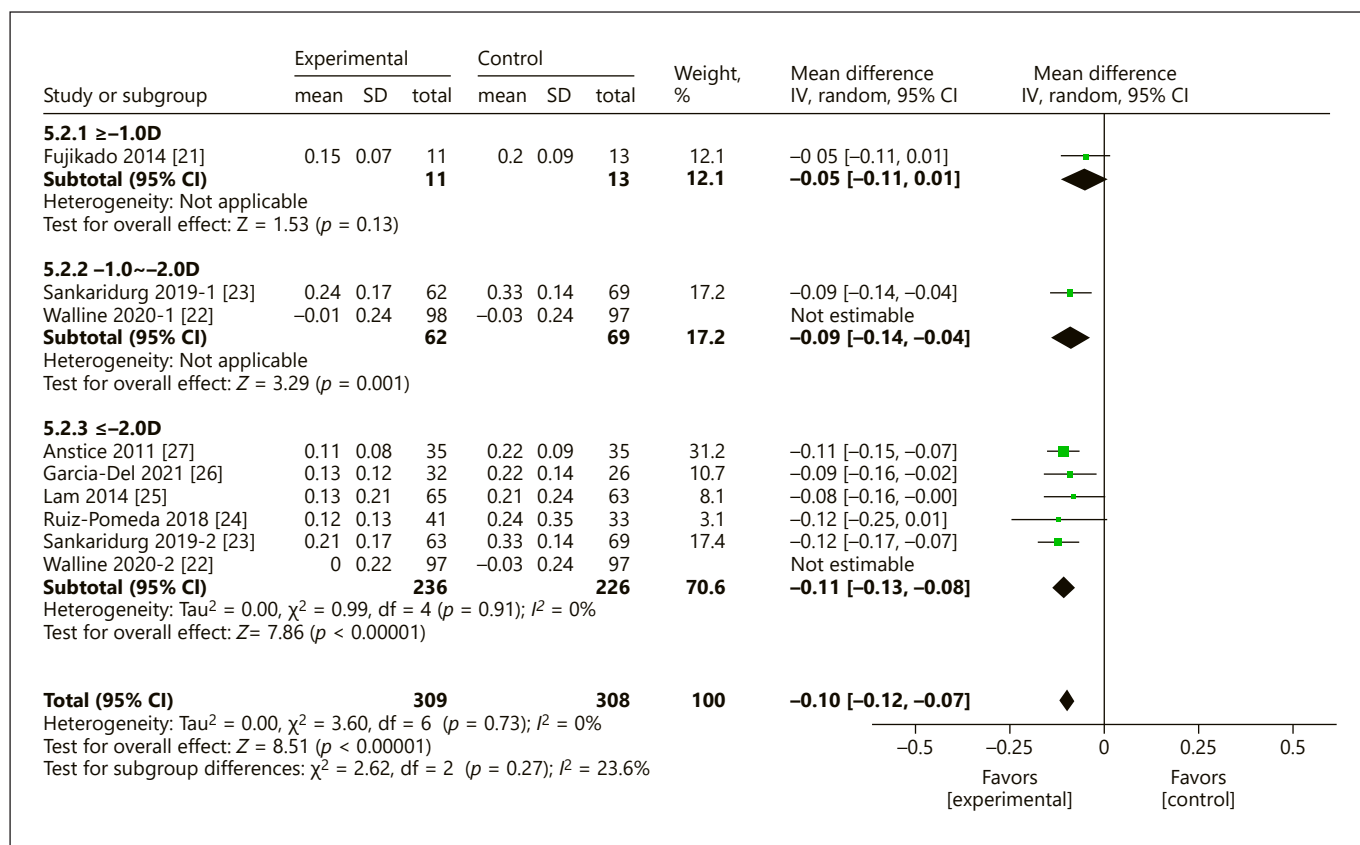


Fig. 9. Forest plots of AL changes in different contact lens additions at 1 year.

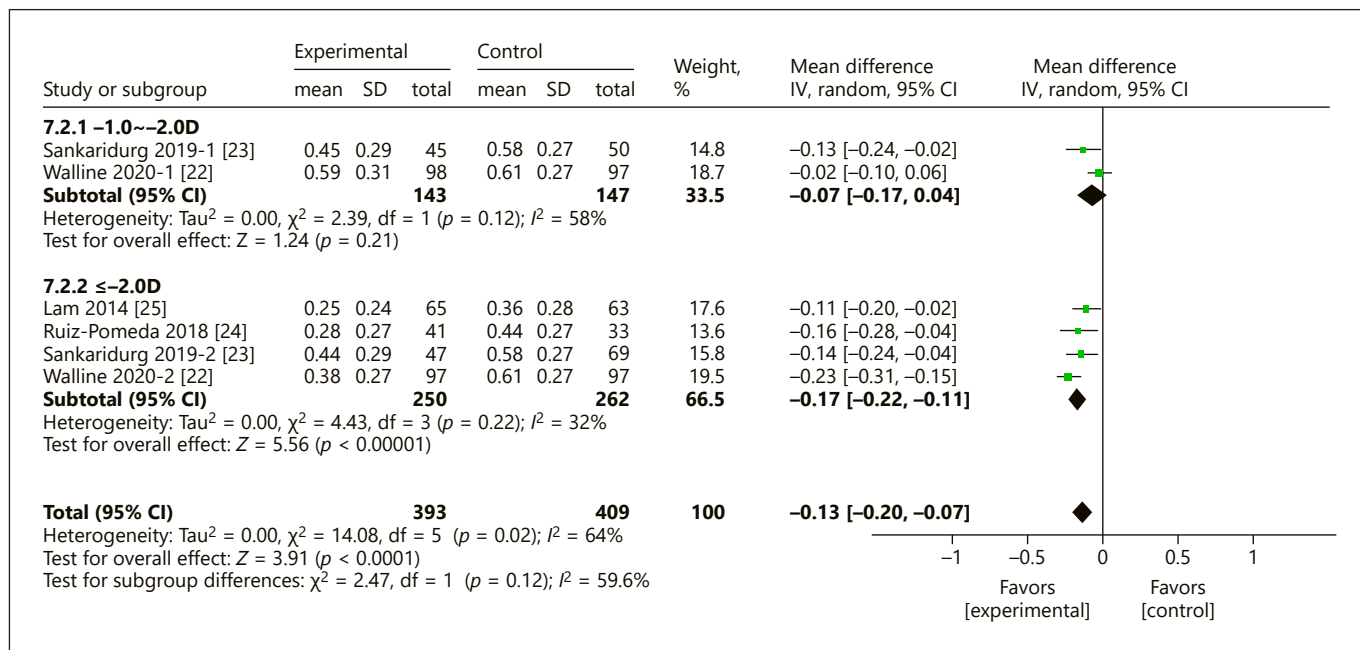


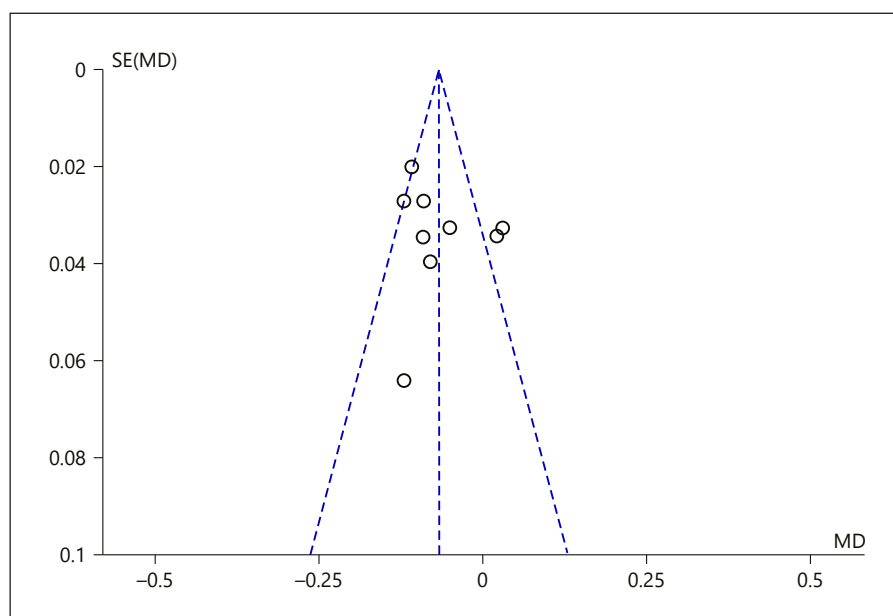
Fig. 10. Forest plots of AL changes in different contact lens additions at 2 years.

Table 4. Sensitivity analysis at 1 year

	Total (95% CI)	Heterogeneity			Test for overall effect	Test of subgroup differences
All studies	−0.07 (−0.10, −0.03)	$\text{Tau}^2 = 0$	$p = 0.002$	$I^2 = 68\%$	$p = 0.0004$	$p = 0.64$
Anstice and Phillips [28] 2011	−0.06 (−0.10, −0.02)	$\text{Tau}^2 = 0$	$p = 0.004$	$I^2 = 66\%$	$p = 0.004$	$p = 0.80$
Fujikado et al. [22] 2014	−0.07 (−0.11, −0.03)	$\text{Tau}^2 = 0$	$p = 0.001$	$I^2 = 71\%$	$p = 0.001$	$p = 0.47$
Garcia-Del1 Valle et al. [27] 202	−0.06 (−0.11, −0.02)	$\text{Tau}^2 = 0$	$p = 0.0008$	$I^2 = 72\%$	$p = 0.002$	$p = 0.72$
Lam et al. [26] 2014	−0.07 (−0.11, −0.02)	$\text{Tau}^2 = 0$	$p = 0.0008$	$I^2 = 72\%$	$p = 0.002$	$p = 0.68$
Ruiz-Pomeda et al. 25] 2018	−0.06 (−0.10, −0.02)	$\text{Tau}^2 = 0$	$p = 0.0009$	$I^2 = 72\%$	$p = 0.001$	$p = 0.73$
Sankaridurg et al. [23] 2019-1	−0.06 (−0.11, −0.02)	$\text{Tau}^2 = 0$	$p = 0.0009$	$I^2 = 72\%$	$p = 0.004$	$p = 0.06$
Sankaridurg et al. [24] 2019-2	−0.06 (−0.10, −0.02)	$\text{Tau}^2 = 0$	$p = 0.003$	$I^2 = 68\%$	$p = 0.004$	$p = 0.82$
Walline et al. [23] 2020	−0.10 (−0.12, −0.07)	$\text{Tau}^2 = 0$	$p = 0.73$	$I^2 = 0\%$	$p < 0.00001$	$p = 0.27$

Table 5. Sensitivity analysis at 2 years

	Total (95% CI)	Heterogeneity			Test for overall effect	Test of subgroup differences
All studies	−0.13 (−0.20, −0.06)	$\text{Tau}^2 = 0$	$p = 0.02$	$I^2 = 64\%$	$p = 0.0001$	$p = 0.12$
Lam et al. [26] 2014	−0.14 (−0.22, −0.05)	$\text{Tau}^2 = 0.01$	$p = 0.008$	$I^2 = 71\%$	$p = 0.001$	$p = 0.05$
Ruiz-Pomeda et al. [25] 2018	−0.13 (−0.20, −0.05)	$\text{Tau}^2 = 0.01$	$p = 0.008$	$I^2 = 71\%$	$p = 0.001$	$p = 0.15$
Sankaridurg et al. [24] 2019-1	−0.13 (−0.21, −0.05)	$\text{Tau}^2 = 0.01$	$p = 0.007$	$I^2 = 72\%$	$p = 0.001$	$p = 0.004$
Sankaridurg et al. [24] 2019-2	−0.13 (−0.21, −0.05)	$\text{Tau}^2 = 0.01$	$p = 0.007$	$I^2 = 72\%$	$p = 0.001$	$p = 0.13$
Walline et al. [23] 2020-1	−0.16 (−0.21, −0.11)	$\text{Tau}^2 = 0$	$p = 0.31$	$I^2 = 17\%$	$p < 0.00001$	$p = 0.58$
Walline et al. [23] 2020-2	−0.10 (−0.15, −0.05)	$\text{Tau}^2 = 0$	$p = 0.24$	$I^2 = 27\%$	$p = 0.0002$	$p = 0.31$

**Fig. 11.** Funnel plot of AL. AL, axial length.

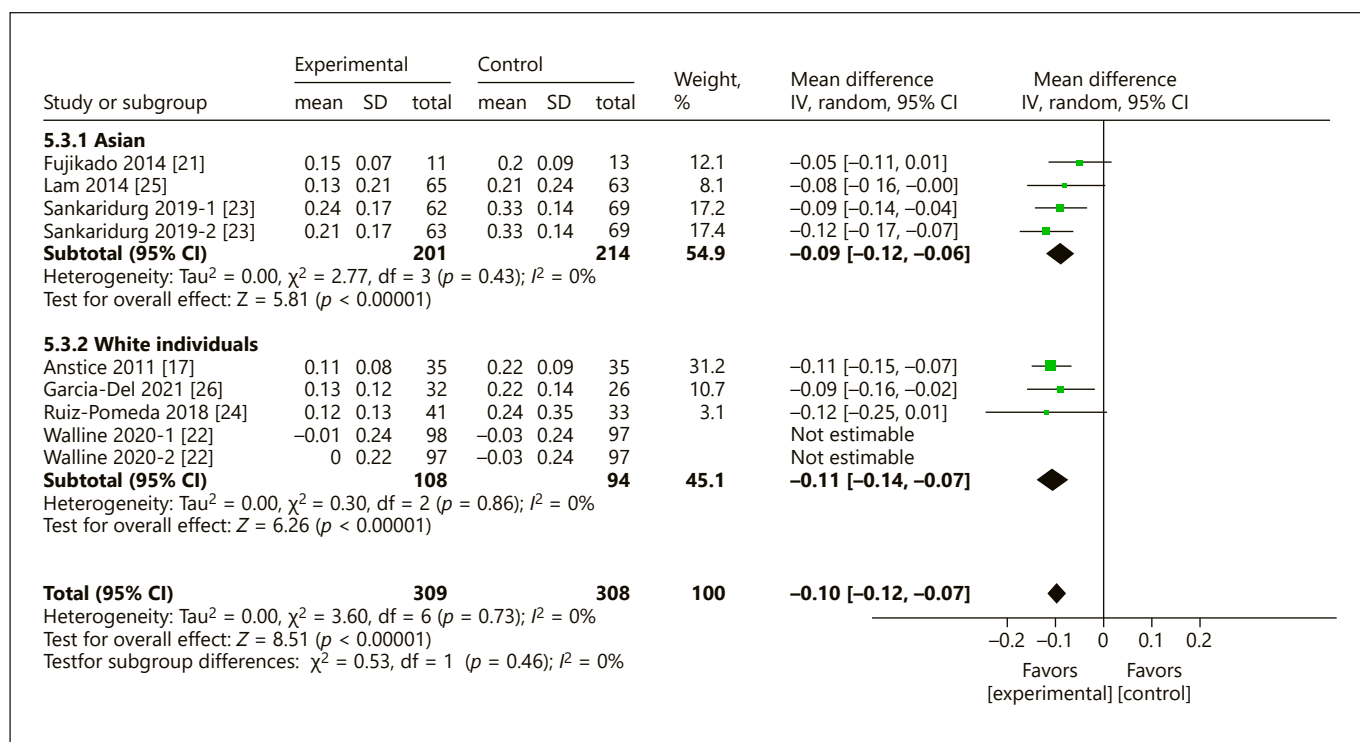


Fig. 12. Forest plots of AL changes in different ethnic groups at 1 year.

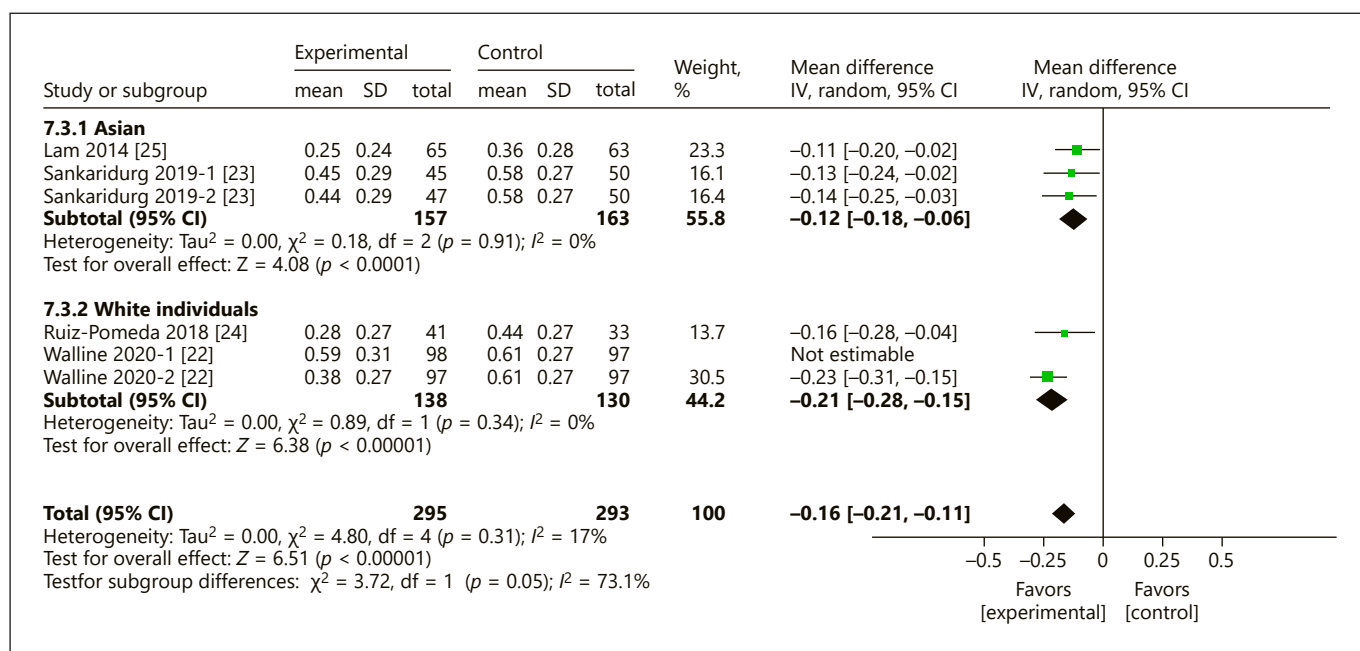


Fig. 13. Forest plots of AL changes in different ethnic groups at 2 years.

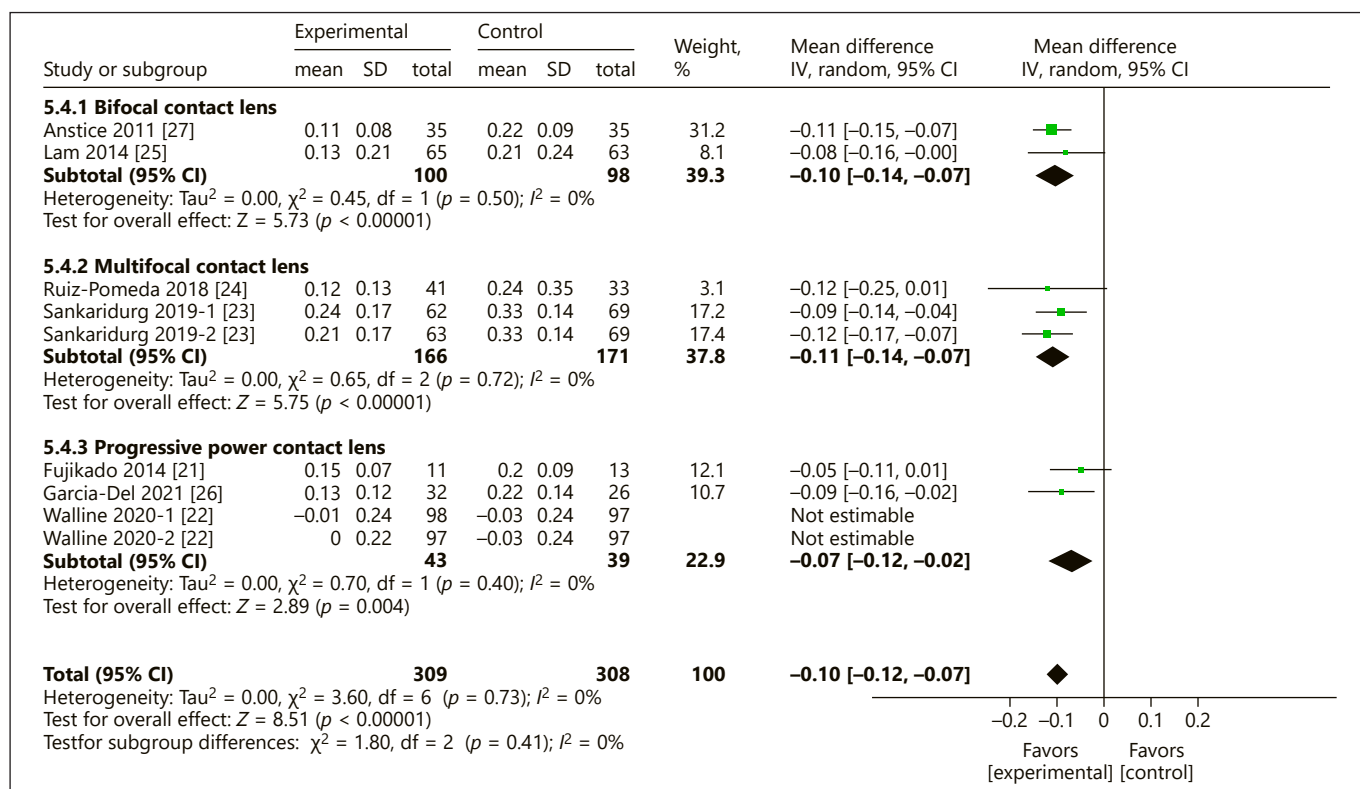


Fig. 14. Forest plots of AL changes in different contact lens design types at 1 year.

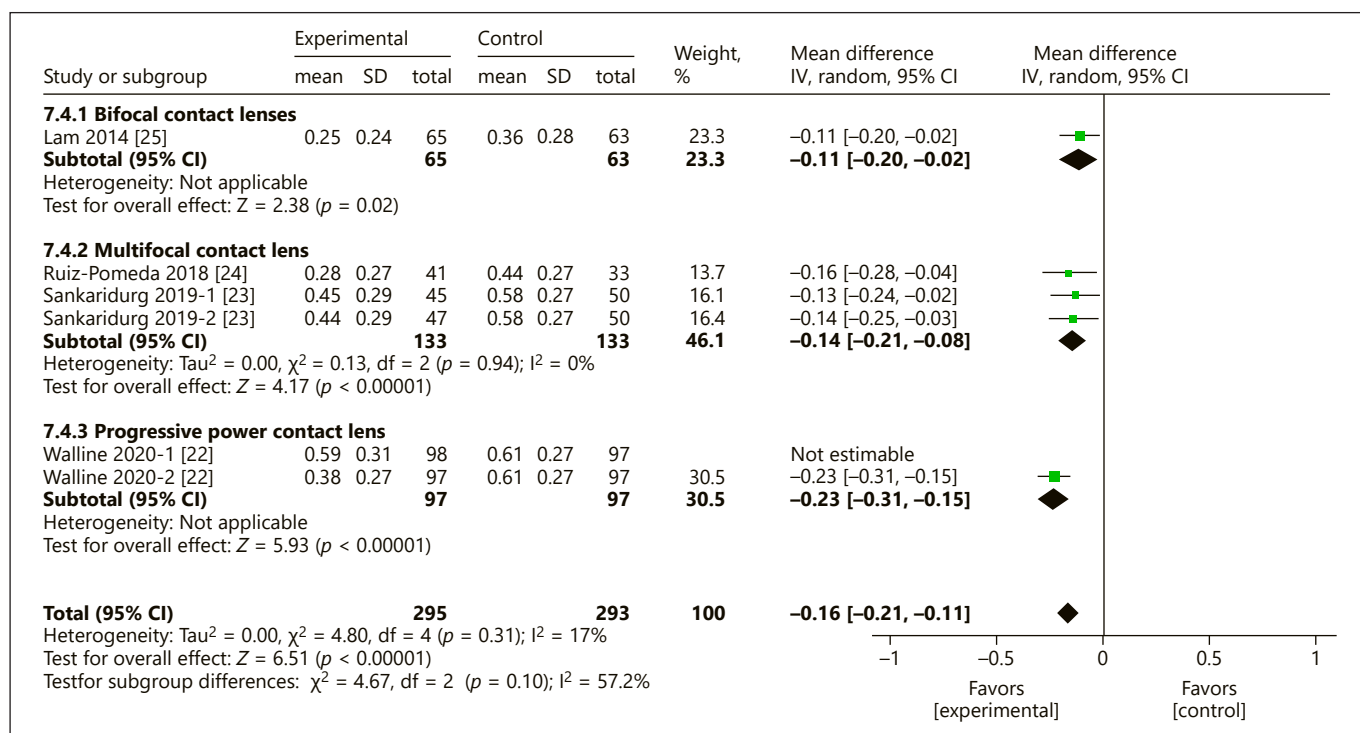


Fig. 15. Forest plots of AL changes in different contact lens design types at 2 years.

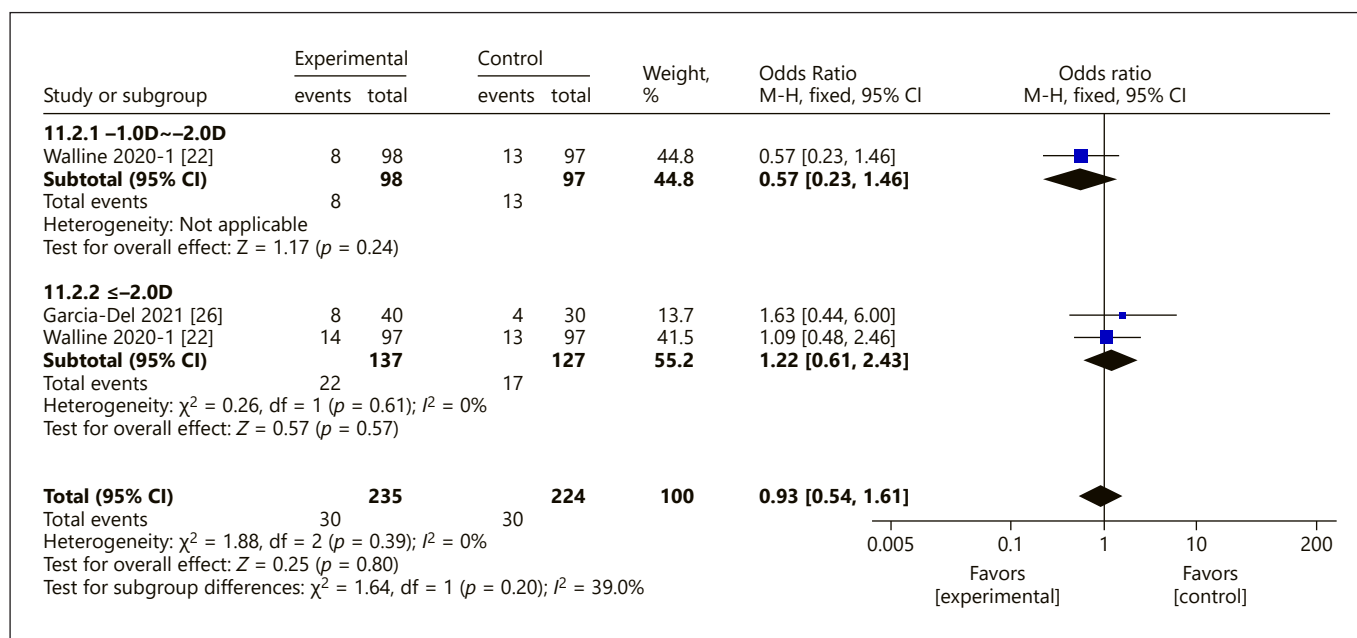


Fig. 16. Forest plot of adverse effects.

Table 6. The adverse events for SMCLs groups versus controls during the treatment of myopia in children

Outcomes	Studies, n	Patients, n
Giant papillary conjunctivitis	2	10
Infiltrative keratitis	1	8
Allergy: ocular	1	7
Micropapillary response	1	4
Corneal neovascularization	1	3
Superficial punctate keratitis	1	3
Corneal epithelial defect/erosion	1	3
Contact lens associated red eye	2	3
Sterile corneal ulcer	1	2
Toxic: gas, soap, etc., in eye	1	2
Superior epithelial arcuate lesion	1	1
Probable microbial keratitis	1	1

reported accommodation, with one presenting a lagged accommodative response in the SMCLs group and the other finding no difference in amplitudes of accommodation between groups.

Acceptability

The dropout rate in the SMCLs and control groups was between 0–55% and 0–49%, respectively. No significant differences regarding dropout rate were detected between

the two groups (odds ratio = 1.08; 95% CI: 0.81–1.45; $p = 0.59$; Fig. 17). There were also no significant differences regarding the dropout rate between different add power SMCLs ($p = 0.74$; Fig. 18).

Discussion

Our current meta-analysis confirmed that SMCLs are effective in slowing myopia progression in children. Differences were found between SMCLs of different add powers, with high add power contact lenses being more effective and stable in controlling myopia progression in children. On the other hand, the medium and low contact lenses showed no myopia control efficacy. No significant differences were observed between the values of axial length change, adverse effects, and dropout rates for different add power contact lenses. The axial length changes were consistent with the refraction trends, but no differences were observed between different add power subgroups. According to our current results, the effect of SMCLs on visual performance needs to be further investigated.

The axial length data revealed heterogeneity, and the sensitivity analysis indicated that Walline's study might have been the source of the heterogeneity. After analyzing Walline's study baseline data, methodology, and results,

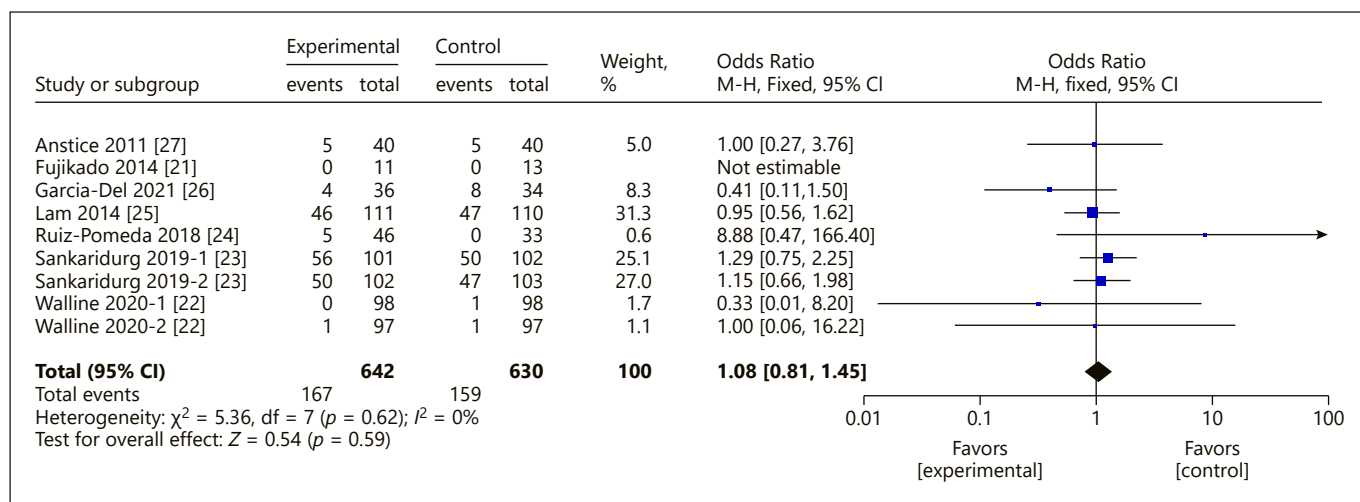


Fig. 17. Forest plot of acceptability in two groups.

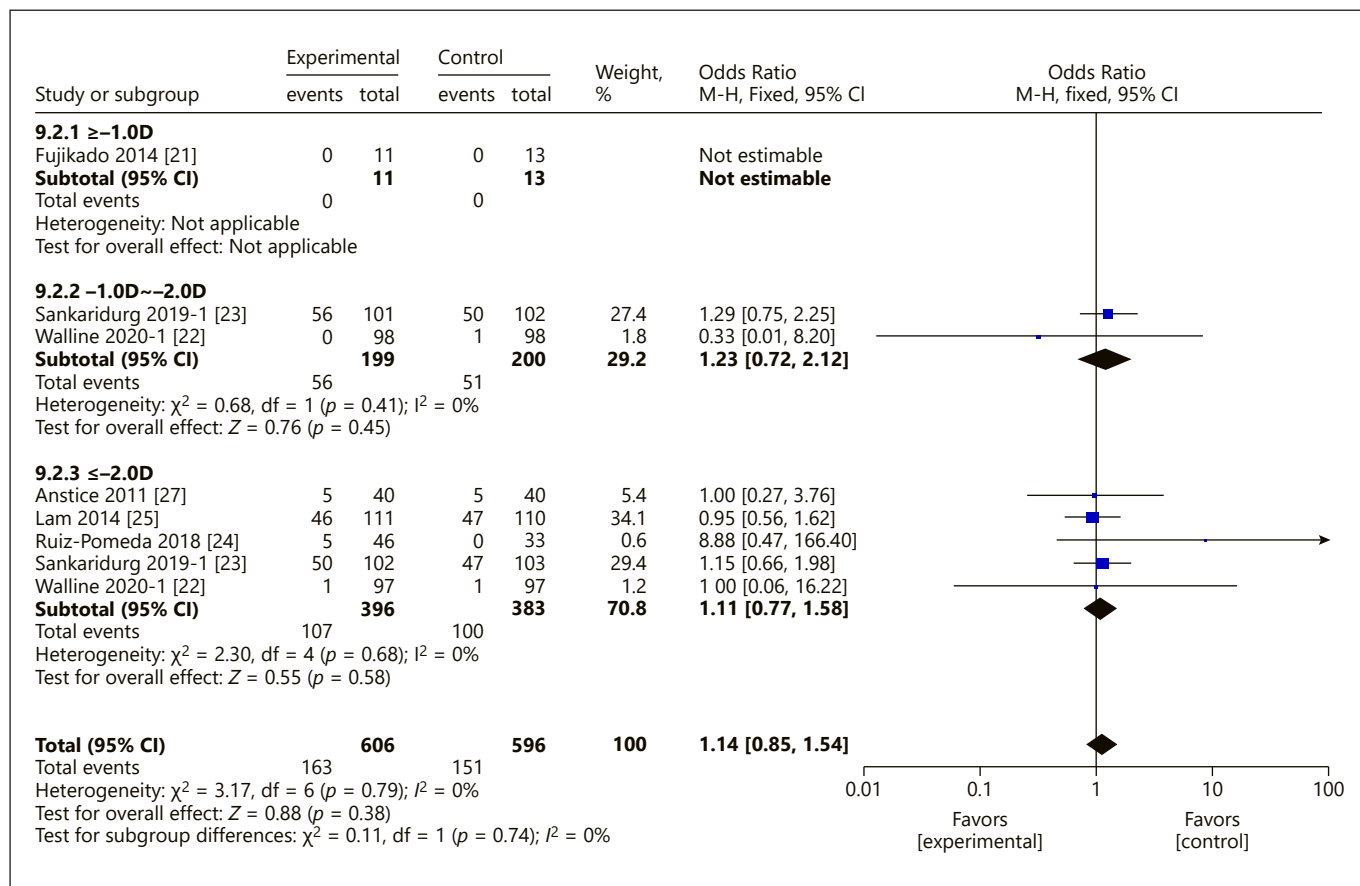


Fig. 18. Forest plot of acceptability in different contact lens additions.

we found that the axial length measurement was performed after ciliary muscle paralysis, whereas several other studies did not. Second, this study had the largest number of completed trials and the smallest rate of loss, whereas other studies had subjects drop out of the trial due to wearing comfort, for example, seeking keratoplasty treatment. Therefore, not including this part of the data would have led to better efficacy. Third, the mean age of the subjects was 10.3 years, while in all other studies they ranged from 11.4 to 14.3 years. Besides, differences in growth and development in children at different ages can affect axial length growth.

A previous meta-study [29] showed that SMCLs were effective in controlling myopia. However, it was not stratified by additive value size and did not quantify adverse effects. Also, this meta-analysis included five RCTs and three cohort studies and did not include medium add power SMCLs. Another meta-analysis [30] confirmed a moderate effect of SMCLs on delaying myopia, and another Bayesian meta-analysis [31] analyzed nine studies and found that SMCLs moderately delayed myopia progression. These meta-analyses were also not stratified by the size of the added power. Another study [32] reviewed only six adverse effects of contact lenses but did not evaluate their efficacy. A systematic review [33] of controlling myopia progression showed that SMCLs were effective. However, only two studies were available at that time. Another meta-analysis [34] evaluating interventions to control myopia progression included only one study of bifocal contact lenses, which showed effectiveness but had a high risk of bias. The results from an updated meta-analysis 9 years later [35], in contrast to the present study, showed that bifocal SCLs were not meaningful for myopia control and did not account for adverse effects. However, only four studies were available at the time. Its results were contrary to our current ones, which might be related to the small number of included studies and its high risk of bias.

Additionally, our quantitative analysis regarding adverse effects helped to form a reference for the selection of the size of the added power for SMCLs. We showed that higher added powers for SMCLs do not lead to more adverse reactions. The incidence of adverse reactions for higher added powers values could not yet be determined. The studies reporting adverse reactions were all in White individuals and progressive refractive design lenses. Therefore, differences in the incidence of adverse reactions regarding different SMCLs designs between Asians and White individuals were not determined due to the limited number of studies.

The small number of studies reporting visual function precluded the collection of pooled data for stratified analysis. Upon systematic evaluation, the main visual-related adverse effects associated with SMCLs were reduced contrast sensitivity and accommodative lag. Indeed, other studies have found that SMCLs can lead to great accommodative lags and accommodation variability [36, 37], reduced peripheral contrast sensitivity [38], and even glares, ghost images [39], and decreased tear film stability [40]. More clinical studies are needed to confirm whether these adverse effects are related to the added power of SMCLs and whether they are dependent on the added power presence. No adverse effects on the anterior cornea, total ocular aberrations [41], and visual quality (high and low contrast distance visual acuity) [42] have been found. Information on efficacy, short- and long-term effects, adverse effects, and risk of vision-related functions require additional examinations to guide clinical intervention decisions.

SMCLs are used to achieve myopia control effects through myopic defocus. Hence, myopic patients look at the near with prolonged peripheral hyperopic defocus, and wearing SMCLs allows the treatment coverage area to be replaced by myopic defocus [43]. Medium add power SMCLs produced little myopic defocus in the periphery [24], and low add power SMCLs did not produce significant myopic defocus [22]. One study found that an add power of +3.00 D was required to invert the peripheral hyperopic defocus pattern to myopic defocus [44]. With different add power values in this range, high add power SMCLs had the best efficacy in controlling myopia in children and did not exhibit significant adverse effects. Although we recommend high add power SMCLs for the control of myopia progression in children, more clinical trials using this added power are required to analyze its adverse effects related to vision. Additionally, more clinical trials with higher add power values are needed to verify whether they have better control, more adverse effects, and more vision-related adverse effects. Although higher add power SMCLs can effectively control myopia progression, myopia control in children and adolescents needs to be combined with outdoor exercise, low concentrations of atropine, and other effective measures.

Limitations

The present study also has some limitations. First, since this study strictly included RCTs, there were not enough studies to investigate the effects of each add power SMCL. Second, due to incomplete reporting of adverse effects, visually relevant adverse effects in the included

studies prevented the determination of the correlation between adverse effects and the added power value, limiting further analysis. Third, studies without a clear added value of SMCLs were excluded and statistical analyses of these studies were missing. Fourth, the efficacy of SMCLs was only reported for 1–2 years and the ability to maintain efficacy was not evaluated. Fifth, SMCLs cause a reduction in visual function in some children and might prompt a reduction in near-eye time, which might be an important factor in slowing myopia progression. This factor was not controlled in any of the studies.

Conclusions

The present meta-analysis showed that, in children, high power-added contact lenses were more effective and stable in controlling myopia progression. Moreover, adverse effects and acceptability were not related to the added power.

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Statement of Ethics

All studies included in the analysis were publicly available and therefore did not require ethical approval and informed consent. However, the study was in full compliance with the Declaration of Helsinki.

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Conflict of Interest Statement

The authors have no conflicts of interest to disclose.

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Author Contributions

Junguo Duan is the guarantor; Zengfang Yu and Junguo Duan contributed to the conception of the study. The manuscript was drafted by Zengfang Yu and Along Zhong and revised by Junguo Duan. The search strategy was developed by all the authors and ran by Xiaoshuang Zhao and Dan Li. Zengfang Yu and Along Zhong also independently screened the studies, extracted data from included studies, assessed the risk of bias, and finished the data synthesis. The disagreement was arbitrated by Xiaoshuang Zhao, who ensured no errors occur during the study. Zengfang Yu and Along Zhong contributed equally to the study.

Data Availability Statement

The data that support the findings of this study are openly available in PubMed at <http://doi.org/10.2147/OPTH.S66884>, reference number [22]; <http://doi.org/10.1001/jama.2020.10834>, reference number [23]; <http://doi.org/10.1111/opo.12621>, reference number [24]; <http://doi.org/10.1007/s00417-018-3906-z>, reference number [25]; <http://doi.org/10.1136/bjophthalmol-2013-303914>, reference number [26]; <http://doi.org/10.1111/cxo.13077>, reference number [27]; <http://doi.org/10.1016/j.ophtha.2010.10.035>, reference number [28]; all data generated or analyzed during this study are included in this article and its online supplementary material files (online suppl. material – data). Further inquiries can be directed to the corresponding author.

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